

CHAPTER 11

SEISMIC DESIGN CRITERIA

11.1 GENERAL

11.1.1 Purpose. Chapter 11 presents criteria for the design and construction of buildings and other structures subject to earthquake ground motions. The specified earthquake loads are based upon postelastic energy dissipation in the structure. Because of this fact, the requirements for design, detailing, and construction shall be satisfied, even for structures and members for which load combinations that do not include earthquake loads indicate larger demands than combinations that include earthquake loads.

11.1.2 Scope. Every structure and portion thereof, including nonstructural components, shall be designed and constructed to resist the effects of earthquake motions as prescribed by the seismic requirements of this standard. Certain nonbuilding structures, as described in Chapter 15, are also within the scope and shall be designed and constructed in accordance with the requirements of Chapter 15. Requirements concerning alterations, additions, and change of use are set forth in Appendix 11B. Existing structures and alterations to existing structures need only comply with the seismic requirements of this standard where required by Appendix 11B. The following structures are exempt from the seismic requirements of this standard:

1. Detached one- and two-family dwellings that are located where the mapped, short period, spectral response acceleration parameter, S_s , is less than 0.4 or where the Seismic Design Category determined in accordance with Section 11.6 is A, B, or C.
2. Detached one- and two-family wood-frame dwellings not included in Exemption 1 with not more than two stories above grade plane, satisfying the limitations of and constructed in accordance with the IRC.
3. Agricultural storage structures that are intended only for incidental human occupancy.
4. Structures that require special consideration of their response characteristics and environment that are not addressed in Chapter 15 and for which other regulations provide seismic criteria, such as vehicular bridges, electrical transmission towers, hydraulic structures, buried utility lines and their appurtenances, and nuclear reactors.
5. Piers and wharves that are not accessible to the general public.

11.1.3 Applicability. Structures and their nonstructural components shall be designed and constructed in accordance with the requirements of the following chapters based on the type of structure or component:

- a. Buildings: Chapter 12;
- b. Nonbuilding Structures: Chapter 15;
- c. Nonstructural Components: Chapter 13;

- d. Seismically Isolated Structures: Chapter 17; and
- e. Structures with Damping Systems: Chapter 18.

Buildings whose purpose is to enclose equipment or machinery and whose occupants are engaged in maintenance or monitoring of that equipment, machinery, or their associated processes shall be permitted to be classified as nonbuilding structures designed and detailed in accordance with Section 15.5 of this standard.

11.1.4 Alternate Materials and Methods of Construction. Alternate materials and methods of construction to those prescribed in the seismic requirements of this standard shall not be used unless approved by the Authority Having Jurisdiction. Substantiating evidence shall be submitted demonstrating that the proposed alternate will be at least equal in strength, durability, and seismic resistance for the purpose intended.

11.1.5 Quality Assurance. Quality assurance for seismic force-resisting systems and other designated seismic systems defined in Section 13.2.2 shall be provided in accordance with the requirements of the Authority Having Jurisdiction.

Where the Authority Having Jurisdiction has not adopted quality assurance requirements, or where the adopted requirements are not applicable to the seismic force-resisting system or designated seismic systems as described in Section 13.2.2, the registered design professional in responsible charge of designing the seismic force-resisting system or other designated seismic systems shall submit a quality assurance plan to the Authority Having Jurisdiction for approval. The quality assurance plan shall specify the quality assurance program elements to be implemented.

11.2 DEFINITIONS

The following definitions apply only to the seismic provisions of Chapters 11 through 22 of this standard.

ACTIVE FAULT: A fault determined to be active by the Authority Having Jurisdiction from properly substantiated data (e.g., most recent mapping of active faults by the U.S. Geological Survey).

ADDITION: An increase in building area, aggregate floor area, height, or number of stories of a structure.

ALTERATION: Any construction or renovation to an existing structure other than an addition.

APPENDAGE: An architectural component such as a canopy, marquee, ornamental balcony, or statuary.

APPROVAL: The written acceptance by the Authority Having Jurisdiction of documentation that establishes the qualification of a material, system, component, procedure, or person to fulfill the requirements of this standard for the intended use.

ATTACHMENTS: Means by which nonstructural components or supports of nonstructural components are secured or connected to the seismic force-resisting system of the structure. Such attachments include anchor bolts, welded connections, and mechanical fasteners.

BASE: The level at which the horizontal seismic ground motions are considered to be imparted to the structure.

BASE SHEAR: Total design lateral force or shear at the base.

BOUNDARY ELEMENTS: Portions along wall and diaphragm edges for transferring or resisting forces. Boundary elements include chords and collectors at diaphragm and shear wall perimeters, edges of openings, discontinuities, and reentrant corners.

BUILDING: Any structure whose intended use includes shelter of human occupants.

CANTILEVERED COLUMN SYSTEM: A seismic force-resisting system in which lateral forces are resisted entirely by columns acting as cantilevers from the base.

CHARACTERISTIC EARTHQUAKE: An earthquake assessed for an active fault having a magnitude equal to the best estimate of the maximum magnitude capable of occurring on the fault but not less than the largest magnitude that has occurred historically on the fault.

COLLECTOR (DRAG STRUT, TIE, DIAPHRAGM STRUT): A diaphragm or shear wall boundary element parallel to the applied load that collects and transfers diaphragm shear forces to the vertical elements of the seismic force-resisting system or distributes forces within the diaphragm or shear wall.

COMPONENT: A part of an architectural, electrical, or mechanical system.

Component, Flexible: Nonstructural component that has a fundamental period greater than 0.06 s.

Component, Nonstructural: A part of an architectural, mechanical, or electrical system within or without a building or nonbuilding structure.

Component, Rigid: Nonstructural component that has a fundamental period less than or equal to 0.06 s.

Component, Rugged: A nonstructural component that has been shown to consistently function after design earthquake level or greater seismic events based on past earthquake experience data or past seismic testing when adequately anchored or supported. The classification of a nonstructural component as rugged shall be based on a comparison of the specific component with components of similar strength and stiffness. Common examples of rugged components include AC motors, compressors, and base-mounted horizontal pumps.

CONCRETE:

Plain Concrete: Concrete that is either unreinforced or contains less reinforcement than the minimum amount specified in ACI 318 for reinforced concrete.

Reinforced Concrete: Concrete reinforced with no less reinforcement than the minimum amount required by ACI 318 prestressed or nonprestressed and designed on the assumption that the two materials act together in resisting forces.

CONSTRUCTION DOCUMENTS: The written, graphic, electronic, and pictorial documents describing the design, locations, and physical characteristics of the project required to verify compliance with this standard.

COUPLING BEAM: A beam that is used to connect adjacent concrete wall elements to make them act together as a unit to resist lateral loads.

DEFORMABILITY: The ratio of the ultimate deformation to the limit deformation.

High-Deformability Element: An element whose deformability is not less than 3.5 where subjected to four fully reversed cycles at the limit deformation.

Limited-Deformability Element: An element that is neither a low-deformability nor a high-deformability element.

Low-Deformability Element: An element whose deformability is 1.5 or less.

DEFORMATION:

Limit Deformation: Two times the initial deformation that occurs at a load equal to 40% of the maximum strength.

Ultimate Deformation: The deformation at which failure occurs and that shall be deemed to occur if the sustainable load reduces to 80% or less of the maximum strength.

DESIGN EARTHQUAKE: The earthquake effects that are two-thirds of the corresponding risk-targeted maximum considered earthquake (MCE_R) effects.

DESIGN EARTHQUAKE GROUND MOTION: The earthquake ground motions that are two-thirds of the corresponding MCE_R ground motions.

DESIGNATED SEISMIC SYSTEMS: Those nonstructural components that require design in accordance with Chapter 13 and for which the component Importance Factor, I_p , is greater than 1.0.

DIAPHRAGM: Roof, floor, or other membrane or bracing system acting to transfer the lateral forces to the vertical resisting elements.

Flexure-Controlled Diaphragm: Diaphragm with a flexural yielding mechanism, which limits the maximum forces that develop in the diaphragm, and having a design shear strength or factored nominal shear capacity greater than the shear corresponding to the nominal flexural strength.

Shear-Controlled Diaphragm: Diaphragm that does not meet the requirements of a flexure-controlled diaphragm.

Transfer Forces, Diaphragm: Forces that occur in a diaphragm caused by transfer of seismic forces from the vertical seismic force-resisting elements above the diaphragm to other vertical seismic force-resisting elements below the diaphragm because of offsets in the placement of the vertical elements or changes in relative lateral stiffnesses of the vertical elements.

Vertical Diaphragm: See WALL, Shear Wall.

DIAPHRAGM BOUNDARY: A location where shear is transferred into or out of the diaphragm element. Transfer is either to a boundary element or to another force-resisting element.

DIAPHRAGM CHORD: A diaphragm boundary element perpendicular to the applied load that is assumed to take axial stresses caused by the diaphragm moment.

DISTRIBUTION SYSTEM: An interconnected system of piping, tubing, conduit, raceway, or duct. Distribution systems include in-line components such as valves, in-line suspended pumps, and mixing boxes.

ELEMENT ACTION: Element axial, shear, or flexural behavior.

Critical Action: An action, failure of which would result in the collapse of multiple bays or multiple stories of the building or would result in a significant reduction in the structure's seismic resistance.

Deformation-Controlled Action: Element actions for which reliable inelastic deformation capacity is achievable without critical strength decay.

Force-Controlled Action: Any element actions modeled with linear properties and element actions not classified as deformation-controlled.

Noncritical Actions: An action, failure of which would not result in either collapse or significant loss of the structure's seismic resistance.

Ordinary Action: An action, failure of which would result in only local collapse, comprising not more than one bay in a single story, and would not result in a significant reduction of the structure's seismic resistance.

ENCLOSURE: An interior space surrounded by walls.

EQUIPMENT SUPPORT: Those structural members or assemblies of members or manufactured elements, including braces, frames, legs, lugs, snuggers, hangers, or saddles, that transmit gravity loads and operating loads between the equipment and the structure.

FLEXIBLE CONNECTIONS: Those connections between equipment components that permit rotational and/or translational movement without degradation of performance. Examples include universal joints, bellows expansion joints, and flexible metal hose.

FOUNDATION GEOTECHNICAL CAPACITY: The maximum pressure or strength design capacity of a foundation based upon the supporting soil, rock, or controlled low-strength material.

FOUNDATION STRUCTURAL CAPACITY: The design strength of foundations or foundation components as provided by adopted material standards and as altered by the requirements of this standard.

FRAME:

Braced Frame: An essentially vertical truss, or its equivalent, of the concentric or eccentric type that is provided in a building frame system or dual system to resist seismic forces.

Concentrically Braced Frame (CBF): A braced frame in which the members are subjected primarily to axial forces. CBFs are categorized as ordinary concentrically braced frames (OCBFs) or special concentrically braced frames (SCBFs).

Eccentrically Braced Frame (EBF): A diagonally braced frame in which at least one end of each brace frames into a beam a short distance from a beam-column or from another diagonal brace.

Moment Frame: A frame in which members and joints resist lateral forces by flexure and along the axis of the members. Moment frames are categorized as intermediate moment frames (IMFs), ordinary moment frames (OMFs), and special moment frames (SMFs).

Structural System:

Building Frame System: A structural system with an essentially complete space frame providing support for vertical loads. Seismic force resistance is provided by shear walls or braced frames.

Dual System: A structural system with an essentially complete space frame providing support for vertical loads. Seismic force resistance is provided by moment-resisting frames and shear walls or braced frames as prescribed in Section 12.2.5.1.

Shear Wall-Frame Interactive System: A structural system that uses combinations of ordinary reinforced concrete shear walls and ordinary reinforced concrete moment frames designed to resist lateral forces in proportion to their rigidities considering interaction between shear walls and frames on all levels.

Space Frame System: A 3-D structural system composed of interconnected members, other than bearing walls, that is capable of supporting vertical loads and, where designed for such an application, is capable of providing resistance to seismic forces.

FRICTION CLIP: A device that relies on friction to resist applied loads in one or more directions to anchor a nonstructural

component. Friction is provided mechanically and is not due to gravity loads.

GLAZED CURTAIN WALL: A nonbearing wall that extends beyond the edges of building floor slabs and includes a glazing material installed in the curtain wall framing.

GLAZED STOREFRONT: A nonbearing wall that is installed between floor slabs, typically including entrances, and includes a glazing material installed in the storefront framing.

GRADE PLANE: A horizontal reference plane representing the average of finished ground level adjoining the structure at all exterior walls. Where the finished ground level slopes away from the exterior walls, the grade plane is established by the lowest points within the area between the structure and the property line or, where the property line is more than 6 ft (1,829 mm) from the structure, between the structure and points 6 ft (1,829 mm) from the structure.

HEATING, VENTILATING, AIR-CONDITIONING, AND REFRIGERATION (HVACR): The equipment, distribution systems, and terminals, excluding interconnecting piping and ductwork that provide, either collectively or individually, the processes of heating, ventilating, air-conditioning, or refrigeration to a building or portion of a building.

INSPECTION, SPECIAL: The observation of the work by a special inspector to determine compliance with the approved construction documents and these standards in accordance with the quality assurance plan.

Continuous Special Inspection: The full-time observation of the work by a special inspector who is present in the area where work is being performed.

Periodic Special Inspection: The part-time or intermittent observation of the work by a special inspector who is present in the area where work has been or is being performed.

INSPECTOR, SPECIAL: A person approved by the Authority Having Jurisdiction to perform special inspection, and who shall be identified as the owner's inspector.

INVERTED PENDULUM-TYPE STRUCTURES: Structures in which more than 50% of the structure's mass is concentrated at the top of a slender, cantilevered structure and in which stability of the mass at the top of the structure relies on rotational restraint to the top of the cantilevered element.

JOINT: The geometric volume common to intersecting members.

LIGHT-FRAME CONSTRUCTION: A method of construction where the structural assemblies (e.g., walls, floors, ceilings, and roofs) are primarily formed by a system of repetitive wood or cold-formed steel framing members or subassemblies of these members (e.g., trusses).

LONGITUDINAL REINFORCEMENT RATIO: Area of longitudinal reinforcement divided by the cross-sectional area of the concrete.

MAXIMUM CONSIDERED EARTHQUAKE (MCE) GROUND MOTION: The most severe earthquake effects considered by this standard, more specifically defined in the following two terms:

Maximum Considered Earthquake Geometric Mean (MCE_G) Peak Ground Acceleration: The most severe earthquake effects considered by this standard determined for geometric mean peak ground acceleration and without adjustment for targeted risk. The MCE_G peak ground acceleration adjusted for site effects (PGA_M) is used in this standard for evaluation of liquefaction, lateral spreading, seismic settlements, and other soil-related issues. In this standard, general procedures for determining PGA_M are provided in Section 11.8.3; site-specific procedures are provided in Section 21.5.

Risk-Targeted Maximum Considered Earthquake (MCE_R)
Ground Motion Response Acceleration: The most severe earthquake effects considered by this standard determined for the orientation that results in the largest maximum response to horizontal ground motions and with adjustment for targeted risk. In this standard, general procedures for determining the MCE_R ground motion values are provided in Section 11.4.4; site-specific procedures are provided in Sections 21.1 and 21.2.

MECHANICALLY ANCHORED TANKS OR VESSELS: Tanks or vessels provided with mechanical anchors to resist overturning moments.

NONBUILDING STRUCTURE: A structure, other than a building, constructed of a type included in Chapter 15 and within the limits of Section 15.1.1.

NONBUILDING STRUCTURE SIMILAR TO A BUILDING: A nonbuilding structure that is designed and constructed in a manner similar to buildings, responds to strong ground motion in a fashion similar to buildings, and has a basic lateral and vertical seismic force-resisting system conforming to one of the types indicated in Tables 12.2-1 or 15.4-1.

OPEN-TOP TANK: A tank without a fixed roof or cover, floating cover, gas holder cover, or dome.

ORTHOGONAL: In two horizontal directions, at 90° to each other.

OWNER: Any person, agent, firm, or corporation that has a legal or equitable interest in a property.

P-DELTA EFFECT: The secondary effect on shears and moments of structural members caused by the action of the vertical loads induced by horizontal displacement of the structure resulting from various loading conditions.

PARTITION: A nonstructural interior wall that spans horizontally or vertically from support to support. The supports may be the basic building frame, subsidiary structural members, or other portions of the partition system.

PILE: Deep foundation element, which includes piers, caissons, and piles.

PILE CAP: Foundation elements to which piles are connected, including grade beams and mats.

PREMANUFACTURED MODULAR MECHANICAL AND ELECTRICAL SYSTEM: A prebuilt, fully or partially enclosed assembly of mechanical and electrical components.

REGISTERED DESIGN PROFESSIONAL: An architect or engineer registered or licensed to practice professional architecture or engineering, as defined by the statutory requirements of the professional registration laws of the state in which the project is to be constructed.

SEISMIC DESIGN CATEGORY: A classification assigned to a structure based on its Risk Category and the severity of the design earthquake ground motion at the site, as defined in Section 11.4.

SEISMIC FORCE-RESISTING SYSTEM: That part of the structural system that has been considered in the design to provide the required resistance to the seismic forces prescribed herein.

SEISMIC FORCES: The assumed forces prescribed herein, related to the response of the structure to earthquake motions, to be used in the design of the structure and its components.

SELF-ANCHORED TANKS OR VESSELS: Tanks or vessels that are stable under design overturning moment without the need for mechanical anchors to resist uplift.

SHEAR PANEL: A floor, roof, or wall element sheathed to act as a shear wall or diaphragm.

SITE CLASS: A classification assigned to a site based on the types of soils present and their engineering properties, as defined in Chapter 20.

STORAGE RACKS, STEEL: A framework or assemblage, comprised of cold-formed or hot-rolled steel structural members, intended for storage of materials, including, but not limited to, pallet storage racks, selective racks, movable-shelf racks, rack-supported systems, automated storage and retrieval systems (stacker racks), push-back racks, pallet-flow racks, case-flow racks, pick modules, and rack-supported platforms. Other types of racks, such as drive-in or drive-through racks, cantilever racks, portable racks, or racks made of materials other than steel, are not considered steel storage racks for the purpose of this standard.

STORAGE RACKS, STEEL CANTILEVERED: A framework or assemblage comprised of cold-formed or hot-rolled steel structural members, primarily in the form of vertical columns, extended bases, horizontal arms projecting from the faces of the columns, and longitudinal (down-aisle) bracing between columns. There may be shelf beams between the arms, depending on the products being stored; this definition does not include other types of racks such as pallet storage racks, drive-in racks, drive-through racks, or racks made of materials other than steel.

STORY: The portion of a structure between the tops of two successive floor surfaces and, for the topmost story, from the top of the floor surface to the top of the roof surface.

STORY ABOVE GRADE PLANE: A story in which the floor or roof surface at the top of the story is more than 6 ft (1,828 mm) above grade plane or is more than 12 ft (3,658 mm) above the finished ground level at any point on the perimeter of the structure.

STORY DRIFT: The horizontal deflection at the top of the story relative to the bottom of the story as determined in Section 12.8.6.

STORY DRIFT RATIO: The story drift, as determined in Section 12.8.6, divided by the story height, h_{sx} .

STORY SHEAR: The summation of design lateral seismic forces at levels above the story under consideration.

STRENGTH:

Design Strength: Nominal strength multiplied by a strength reduction factor, ϕ .

Nominal Strength: Strength of a member or cross section calculated in accordance with the requirements and assumptions of the strength design methods of this standard (or the reference documents) before application of any strength-reduction factors.

Required Strength: Strength of a member, cross section, or connection required to resist factored loads or related internal moments and forces in such combinations as stipulated by this standard.

STRUCTURAL HEIGHT: The vertical distance from the base to the highest level of the seismic force-resisting system of the structure. For pitched or sloped roofs, the structural height is from the base to the average height of the roof.

STRUCTURAL OBSERVATIONS: The visual observations to determine that the seismic force-resisting system is constructed in general conformance with the construction documents.

STRUCTURE: That which is built or constructed and limited to buildings and nonbuilding structures as defined herein.

SUBDIAPHRAGM: A portion of a diaphragm used to transfer wall anchorage forces to diaphragm cross-ties.

SUPPORTS: Those members, assemblies of members, or manufactured elements, including braces, frames, legs, lugs, snubbers, hangers, saddles, or struts, and associated fasteners that transmit loads between nonstructural components and their attachments to the structure.

TESTING AGENCY: A company or corporation that provides testing and/or inspection services.

VENEERS: Facings or ornamentation of brick, concrete, stone, tile, or similar materials attached to a backing.

WALL: A component that has a slope of 60 deg or greater with the horizontal plane used to enclose or divide space.

Bearing Wall: Any wall meeting either of the following classifications:

1. Any metal or wood stud wall that supports more than 100 lb/linear ft (1,459 N/m) of vertical load in addition to its own weight.
2. Any concrete or masonry wall that supports more than 200 lb/linear ft (2,919 N/m) of vertical load in addition to its own weight.

Light Frame Wall: A wall with wood or steel studs.

Light Frame Wood Shear Wall: A wall constructed with wood studs and sheathed with material rated for shear resistance.

Nonbearing Wall: Any wall that is not a bearing wall.

Nonstructural Wall: A wall other than a bearing wall or shear wall.

Shear Wall (Vertical Diaphragm): A wall, bearing or non-bearing, designed to resist lateral forces acting in the plane of the wall (sometimes referred to as a “vertical diaphragm”).

Structural Wall: A wall that meets the definition for bearing wall or shear wall.

WALL SYSTEM, BEARING: A structural system with bearing walls providing support for all or major portions of the vertical loads. Shear walls or braced frames provide seismic force resistance.

WOOD STRUCTURAL PANEL: A wood-based panel product that meets the requirements of DOC PS1 or DOC PS2 and is bonded with a waterproof adhesive. Included under this designation are plywood, oriented strand board, and composite panels.

11.3 SYMBOLS

The unit dimensions used with the items covered by the symbols shall be consistent throughout except where specifically noted. Symbols presented in this section apply only to the seismic provisions of Chapters 11 through 22 in this standard.

- A_0 = area of the load-carrying foundation [ft² (m²)]
- A_{ch} = cross-sectional area [in.² (mm²)] of a structural member measured out-to-out of transverse reinforcement
- A_{sh} = total cross-sectional area of hoop reinforcement [in.² (mm²)], including supplementary crossties, having a spacing of s_h and crossing a section with a core dimension of h_c
- A_{vd} = required area of leg [in.² (mm²)] of diagonal reinforcement
- A_x = torsional amplification factor (Section 12.8.4.3)
- a_i = the acceleration at level i obtained from a modal analysis (Section 13.3.1)
- a_p = the amplification factor related to the response of a system or component as affected by the type of seismic attachment, determined in Section 13.3.1
- b_p = the width of the rectangular glass panel
- C_d = deflection amplification factor as given in Tables 12.2-1, 15.4-1, or 15.4-2
- C_{dX} = deflection amplification factor in the X direction (Section 12.9.2.5)
- C_{dY} = deflection amplification factor in the Y direction (Section 12.9.2.5)

- C_{p0} = diaphragm design acceleration coefficient at the structure base (Section 12.10.3.2.1)
- C_{pi} = diaphragm design acceleration coefficient at 80% of the structural height above the base, h_n (Section 12.10.3.2.1)
- C_{pn} = diaphragm design acceleration coefficient at the structural height, h_n (Section 12.10.3.2.1)
- C_{px} = diaphragm design acceleration coefficient at level x (Section 12.10.3.2.1)
- C_R = site-specific risk coefficient at any period (Section 21.2.1.1)
- C_{R1} = mapped value of the risk coefficient at a period of 1 s as given by Fig. 22-19
- C_{RS} = mapped value of the risk coefficient at short periods as given by Fig. 22-18
- C_s = seismic response coefficient determined in Section 12.8.1.1 or 19.3.1 (dimensionless)
- C_{s2} = higher mode seismic response coefficient (Section 12.10.3.2.1)
- C_t = building period coefficient (Section 12.8.2.1)
- C_{vx} = vertical distribution factor as determined (Section 12.8.3)
- c = distance from the neutral axis of a flexural member to the fiber of maximum compressive strain [in. (mm)]
- D = the effect of dead load
- D_{clear} = relative horizontal (drift) displacement, measured over the height of the glass panel under consideration, which causes initial glass-to-frame contact. For rectangular glass panels within a rectangular wall frame, D_{clear} is set forth in Section 13.5.9.1
- D_{pl} = seismic relative displacement; see Section 13.3.2
- D_s = the total depth of stratum in Eq. (19.3-4) [ft (m)]
- d_c = the total thickness of cohesive soil layers in the top 100 ft (30 m); see Section 20.4.3 [ft (m)]
- d_i = the thickness of any soil or rock layer i [between 0 and 100 ft (between 0 and 30 m)]; see Section 20.4.1 [ft (m)]
- d_s = the total thickness of cohesionless soil layers in the top 100 ft (30 m); see Section 20.4.2 [ft (m)]
- E = effect of horizontal and vertical earthquake-induced forces (Section 12.4)
- E_{cl} = The capacity-limited horizontal seismic load effect, equal to the maximum force that can develop in the element as determined by a rational, plastic mechanism analysis
- F_a = short-period site coefficient (at 0.2-s period); see Section 11.4.4
- F_i, F_n, F_x = portion of the seismic base shear, V , induced at level i, n , or x , respectively, as determined in Section 12.8.3
- F_p = the seismic force acting on a component of a structure as determined in Sections 12.11.1 and 13.3.1
- F_{px} = diaphragm seismic design force at Level x
- F_{PGA} = site coefficient for peak ground acceleration (PGA); see Section 11.8.3
- F_v = long-period site coefficient (at 1.0-s period); see Section 11.4.4
- f'_c = specified compressive strength of concrete used in design
- f'_s = ultimate tensile strength [psi (MPa)] of the bolt, stud, or insert leg wires. For ASTM A307 bolts or ASTM A108 studs, it is permitted to be assumed to be 60,000 psi (415 MPa)

f_y = specified yield strength of reinforcement [psi (MPa)]
 f_{yh} = specified yield strength of the special lateral reinforcement [psi (kPa)]
 $G = \gamma v_s^2 / g$ = the average shear modulus for the soils beneath the foundation at large strain levels [psf (Pa)]
 $G_0 = \gamma v_{s0}^2 / g$ = the average shear modulus for the soils beneath the foundation at small strain levels [psf (Pa)]
 g = acceleration due to gravity
 H = thickness of soil
 h = height of a shear wall measured as the maximum clear height from top of foundation to bottom of diaphragm framing above, or the maximum clear height from top of diaphragm to bottom of diaphragm framing above
 h = average roof height of structure with respect to the base; see Chapter 13
 h^* = effective height of the building as determined in Chapter 19 [ft (m)]
 h_c = core dimension of a component measured to the outside of the special lateral reinforcement [in. (mm)]
 h_i, h_x = the height above the base to level i or x , respectively
 h_n = structural height as defined in Section 11.2
 h_p = the height of the rectangular glass panel
 h_{sx} = the story height below level $x = (h_x - h_{x-1})$
 I_e = the Importance Factor as prescribed in Section 11.5.1
 I_p = the component importance factor as prescribed in Section 13.3.1
 i = the building level referred to by the subscript i ; $i = 1$ designates the first level above the base
 K_p = the stiffness of the component or attachment (Section 13.3.3)
 K_{xx}, K_{rr} = rotational foundation stiffness [Eqs. (19.3-9) and (19.3-19) [ft-lb/degree (N-m/rad)]]
 K_y, K_r = translational foundational stiffness [Eqs. (19.3-8) and (19.3-18)] [lb/in. (N/m)]
 KL/r = the lateral slenderness ratio of a compression member measured in terms of its effective length, KL , and the least radius of gyration of the member cross section, r
 k = distribution exponent given in Section 12.8.3
 k_a = coefficient defined in Sections 12.11.2.1 and 12.14.7.5
 L = overall length of the building (ft or m) at the base in the direction being analyzed
 M_t = torsional moment resulting from eccentricity between the locations of center of mass and the center of rigidity (Section 12.8.4.1)
 M_{ta} = accidental torsional moment as determined in Section 12.8.4.2
 m = a subscript denoting the mode of vibration under consideration; that is, $m = 1$ for the fundamental mode
 N = standard penetration resistance, ASTM D1586
 N = number of stories above the base (Section 12.8.2.1)
 $\bar{N}$ = average field standard penetration resistance for the top 100 ft (30 m); see Sections 20.3.3 and 20.4.2
 $\bar{N}_{ch}$ = average standard penetration resistance for cohesionless soil layers for the top 100 ft (30 m); see Sections 20.3.3 and 20.4.2
 N_i = standard penetration resistance of any soil or rock layer i [between 0 and 100 ft (between 0 and 30 m)]; see Section 20.4.2

n = designation for the level that is uppermost in the main portion of the building
 PGA = mapped MCE_G peak ground acceleration shown in Figs. 22-9 through 22-13
 PGA_M = MCE_G peak ground acceleration adjusted for site class effects; see Section 11.8.3
 PI = plasticity index, ASTM D4318
 P_x = total unfactored vertical design load at and above level x , for use in Section 12.8.7
 Q_E = effect of horizontal seismic (earthquake-induced) forces
 R = response modification coefficient as given in Tables 12.2-1, 12.14-1, 15.4-1, and 15.4-2
 R_p = component response modification factor as defined in Section 13.3.1
 R_s = diaphragm design force reduction factor (Section 12.10.3.5)
 R_X = response modification coefficient in the X direction (Section 12.9.2.5)
 R_Y = response modification coefficient in the Y direction (Section 12.9.2.5)
 S_1 = mapped MCE_R , 5% damped, spectral response acceleration parameter at a period of 1 s as defined in Section 11.4.2
 S_{aM} = the site-specific MCE_R spectral response acceleration parameter at any period
 S_{D1} = design, 5% damped, spectral response acceleration parameter at a period of 1 s as defined in Section 11.4.5
 S_{DS} = design, 5% damped, spectral response acceleration parameter at short periods as defined in Section 11.4.5
 S_{M1} = the MCE_R , 5% damped, spectral response acceleration parameter at a period of 1 s adjusted for site class effects as defined in Section 11.4.4
 S_{MS} = the MCE_R , 5% damped, spectral response acceleration parameter at short periods adjusted for site class effects as defined in Section 11.4.4
 S_S = mapped MCE_R , 5% damped, spectral response acceleration parameter at short periods as defined in Sections 11.4.2, 11.4.4
 s_h = spacing of special lateral reinforcement [in. (mm)]
 s_u = undrained shear strength; see Section 20.4.3
 $\bar{s}_u$ = average undrained shear strength in top 100 ft (30 m); see Sections 20.3.3 and 20.4.3, ASTM D2166, or ASTM D2850
 s_{ui} = undrained shear strength of any cohesive soil layer i [between 0 and 100 ft (0 and 30 m)]; see Section 20.4.3
 T = the fundamental period of the building
 $T_0 = 0.2S_{D1}/S_{DS}$
 $\bar{T}$ = the fundamental period as determined in Chapter 19
 T_a = approximate fundamental period of the building as determined in Section 12.8.2
 T_L = long-period transition period as defined in Section 11.4.6
 T_{lower} = period of vibration at which 90% of the actual mass has been recovered in each of the two orthogonal directions of response (Section 12.9.2). The mathematical model used to compute T_{lower} shall not include accidental torsion and shall include P-delta effects.
 T_p = fundamental period of the component and its attachment (Section 13.3.3)
 $T_{\bar{S}} = S_{D1}/S_{DS}$

- T_{upper} = the larger of the two orthogonal fundamental periods of vibration (Section 12.9.2). The mathematical model used to compute T_{upper} shall not include accidental torsion and shall include P-delta effects
- V = total design lateral force or shear at the base
- V_{EX} = maximum absolute value of elastic base shear computed in the X direction among all three analyses performed in that direction (Section 12.9.2.5)
- V_{EY} = maximum absolute value of elastic base shear computed in the Y direction among all three analyses performed in that direction (Section 12.9.2.5)
- V_{IX} = inelastic base shear in the X direction (Section 12.9.2.5)
- V_{IY} = inelastic base shear in the Y direction (Section 12.9.2.5)
- V_I = design value of the seismic base shear as determined in Section 12.9.1.4.1
- V_X = ELF base shear in the X direction (Section 12.9.2.5)
- V_x = seismic design shear in story x as determined in Section 12.8.4
- V_Y = ELF base shear in the Y direction (Section 12.9.2.5)
- $\tilde{V}$ = reduced base shear accounting for the effects of soil structure interaction as determined in Section 19.3.1
- $\tilde{V}_1$ = portion of the reduced base shear, $\tilde{V}_1$ contributed by the fundamental mode, Section 19.3, in kip (kN)
- ΔV = reduction in V as determined in Section 19.3.1, in kip (kN)
- ΔV_1 = reduction in V_1 as determined in Section 19.3.1, in kip (kN)
- v_s = shear wave velocity at small shear strains (greater than $10^{-3}\%$ strain); see Section 19.2.1, in ft/s (m/s)
- $\bar{v}_s$ = average shear wave velocity at small shear strains in top 100 ft (30 m); see Sections 20.3.3 and 20.4.1
- v_{si} = the shear wave velocity of any soil or rock layer i (between 0 and 100 ft (between 0 and 30 m)); see Section 20.4.1
- v_{so} = average shear wave velocity for the soils beneath the foundation at small strain levels, Section 19.2.1.1 in ft/s (m/s)
- W = effective seismic weight of the building as defined in Section 12.7.2. For calculation of seismic-isolated building period, W is the total effective seismic weight of the building as defined in Sections 19.2 and 19.3, in kip (kN)
- W = effective seismic weight of the building as defined in Sections 19.2 and 19.3, in kip (kN)
- W_c = gravity load of a component of the building
- W_p = component operating weight, in lb (N)
- w_{px} = weight tributary to the diaphragm at level x
- w = moisture content (in percent), ASTM D2216
- w_i, w_n, w_x = portion of W that is located at or assigned to level i , n , or x , respectively
- x = level under consideration, 1 designates the first level above the base
- z = height in structure of point of attachment of component with respect to the base; see Section 13.3.1
- z_s = mode shape factor, Section 12.10.3.2.1
- β = ratio of shear demand to shear capacity for the story between levels x and $x - 1$
- $\bar{\beta}$ = fraction of critical damping for the coupled structure–foundation system, determined in Section 19.2.1
- β_0 = foundation damping factor as specified in Section 19.2.1.2
- Γ_{m1}, Γ_{m2} = first and higher modal contribution factors, respectively, Section 12.10.3.2.1
- γ = average unit weight of soil, in lb/ft³ (N/m³)
- Δ = design story drift as determined in Section 12.8.6
- $\Delta_{fallout}$ = the relative seismic displacement (drift) at which glass fallout from the curtain wall, storefront, or partition occurs
- Δ_a = allowable story drift as specified in Section 12.12.1
- Δ_{ADVE} = average drift of adjoining vertical elements of the seismic force-resisting system over the story below the diaphragm under consideration, under tributary lateral load equivalent to that used in the computation of δ_{MDD} Fig. 12.3-1, in in. (mm)
- δ_{MDD} = computed maximum in-plane deflection of the diaphragm under lateral load, Fig. 12.3-1, in in. (mm)
- δ_{max} = maximum displacement at level x , considering torsion, Section 12.8.4.3
- δ_M = maximum inelastic response displacement, considering torsion, Section 12.12.3
- δ_{MT} = total separation distance between adjacent structures on the same property, Section 12.12.3
- δ_{avg} = the average of the displacements at the extreme points of the structure at level x , Section 12.8.4.3
- δ_x = deflection of level x at the center of the mass at and above level x , Eq. (12.8-15)
- δ_{xc} = deflection of level x at the center of the mass at and above level x determined by an elastic analysis, Section 12.8.6
- δ_{xm} = modal deflection of level x at the center of the mass at and above level x as determined by Section 19.3.2
- $\bar{\delta}_x, \bar{\delta}_{x1}$ = deflection of level x at the center of the mass at and above level x , Eqs. (19.2-13) and (19.3-3), in in. (mm)
- θ = stability coefficient for P-delta effects as determined in Section 12.8.7
- η_x = Force scale factor in the X direction (12.9.2.5)
- η_y = Force scale factor in the Y direction (12.9.2.5)
- ρ = a redundancy factor based on the extent of structural redundancy present in a building as defined in Section 12.3.4
- ρ_s = spiral reinforcement ratio for precast, prestressed piles in Section 14.2.3.2.6
- λ = time effect factor
- Ω_0 = overstrength factor as defined in Tables 12.2-1, 15.4.-1, and 15.4-2
- Ω_v = Diaphragm shear overstrength factor (Section 14.2.4.1.3)

11.4 SEISMIC GROUND MOTION VALUES

11.4.1 Near-Fault Sites. Sites satisfying either of the following conditions shall be classified as near fault:

1. 9.5 miles (15 km) of the surface projection of a known active fault capable of producing $M_w 7$ or larger events, or
2. 6.25 miles (10 km) of the surface projection of a known active fault capable of producing $M_w 6$ or larger events.

EXCEPTIONS:

1. Faults with estimated slip rate along the fault less than 0.04 in. (1 mm) per year shall not be considered.
2. The surface projection shall not include portions of the fault at depths of 6.25 mi (10 km) or greater.

11.4.2 Mapped Acceleration Parameters. The parameters S_S and S_1 shall be determined from the 0.2- and 1-s spectral response accelerations shown in Figs. 22-1, 22-3, 22-5, 22-6, 22-7, and 22-8 for S_S and Figs. 22-2, 22-4, 22-5, 22-6, 22-7, and 22-8 for S_1 . Where S_1 is less than or equal to 0.04 and S_S is less than or equal to

0.15, the structure is permitted to be assigned to Seismic Design Category A and is only required to comply with Section 11.7.

User Note: Electronic values of mapped acceleration parameters and other seismic design parameters are provided at the U.S. Geological Survey (USGS) website at <https://doi.org/10.5066/F7NK3C76>.

11.4.3 Site Class. Based on the site soil properties, the site shall be classified as Site Class A, B, C, D, E, or F in accordance with Chapter 20. Where the soil properties are not known in sufficient detail to determine the site class, Site Class D, subject to the requirements of Section 11.4.4, shall be used unless the authority having jurisdiction or geotechnical data determine that Site Class E or F soils are present at the site.

For situations in which site investigations, performed in accordance with Chapter 20, reveal rock conditions consistent with Site Class B, but site-specific velocity measurements are not made, the site coefficients F_a , F_v , and F_{PGA} shall be taken as unity (1.0).

11.4.4 Site Coefficients and Risk-Targeted Maximum Considered Earthquake (MCE_R) Spectral Response Acceleration Parameters. The MCE_R spectral response acceleration parameters for short periods (S_{MS}) and at 1 s (S_{M1}), adjusted for site class effects, shall be determined by Eqs. (11.4-1) and (11.4-2), respectively.

$$S_{MS} = F_a S_S \tag{11.4-1}$$

$$S_{M1} = F_v S_1 \tag{11.4-2}$$

where

S_S = the mapped MCE_R spectral response acceleration parameter at short periods as determined in accordance with Section 11.4.2, and

S_1 = the mapped MCE_R spectral response acceleration parameter at a period of 1 s as determined in accordance with Section 11.4.2

where site coefficients F_a and F_v are defined in Tables 11.4-1 and 11.4-2, respectively. Where Site Class D is selected as the default site class per Section 11.4.3, the value of F_a shall not be less than 1.2. Where the simplified design procedure of Section 12.14 is used, the value of F_a shall be determined in accordance with Section 12.14.8.1, and the values for F_v , S_{MS} , and S_{M1} need not be determined.

Table 11.4-1 Short-Period Site Coefficient, F_a

Mapped Risk-Targeted Maximum Considered Earthquake (MCE _R) Spectral Response Acceleration Parameter at Short Period						
Site Class	$S_S \leq 0.25$	$S_S = 0.5$	$S_S = 0.75$	$S_S = 1.0$	$S_S = 1.25$	$S_S \geq 1.5$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.9	0.9	0.9	0.9	0.9	0.9
C	1.3	1.3	1.2	1.2	1.2	1.2
D	1.6	1.4	1.2	1.1	1.0	1.0
E	2.4	1.7	1.3	See	See	See
F	See	See	See	See	See	See
	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_S .

Table 11.4-2 Long-Period Site Coefficient, F_v

Mapped Risk-Targeted Maximum Considered Earthquake (MCE _R) Spectral Response Acceleration Parameter at 1-s Period						
Site Class	$S_1 \leq 0.1$	$S_1 = 0.2$	$S_1 = 0.3$	$S_1 = 0.4$	$S_1 = 0.5$	$S_1 \geq 0.6$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.8	0.8	0.8	0.8	0.8	0.8
C	1.5	1.5	1.5	1.5	1.5	1.4
D	2.4	2.2 ^a	2.0 ^a	1.9 ^a	1.8 ^a	1.7 ^a
E	4.2	See	See	See	See	See
F	See	See	See	See	See	See
	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8	Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_1 .
^aAlso, see requirements for site-specific ground motions in Section 11.4.8.

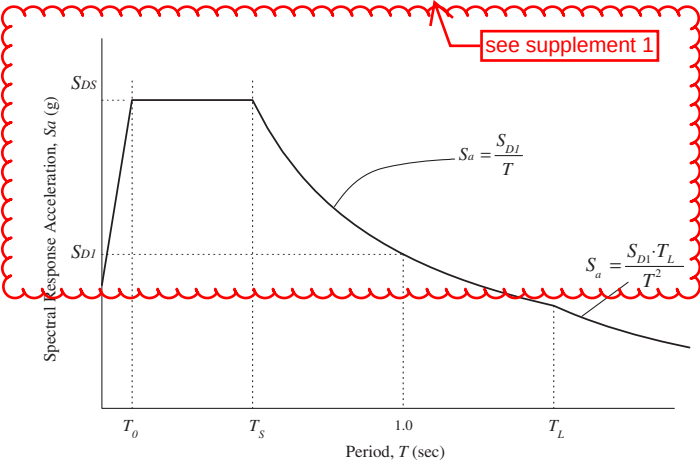


FIGURE 11.4-1 Design Response Spectrum

11.4.5 Design Spectral Acceleration Parameters. Design earthquake spectral response acceleration parameters at short periods, S_{DS} , and at 1-s periods, S_{D1} , shall be determined from Eqs. (11.4-3) and (11.4-4), respectively. Where the alternate simplified design procedure of Section 12.14 is used, the value of S_{DS} shall be determined in accordance with Section 12.14.8.1, and the value for S_{D1} need not be determined.

$$S_{DS} = \frac{2}{3} S_{MS} \tag{11.4-3}$$

$$S_{D1} = \frac{2}{3} S_{M1} \tag{11.4-4}$$

11.4.6 Design Response Spectrum. Where a design response spectrum is required by this standard and site-specific ground motion procedures are not used, the design response spectrum curve shall be developed as indicated in Fig. 11.4-1 and as follows:

1. For periods less than T_0 , the design spectral response acceleration, S_a , shall be taken as given in Eq. (11.4-5):

$$S_a = S_{DS} \left(0.4 + 0.6 \frac{T}{T_0} \right) \tag{11.4-5}$$

- For periods greater than or equal to T_0 and less than or equal to T_S , the design spectral response acceleration, S_a , shall be taken as equal to S_{DS} .
- For periods greater than T_S and less than or equal to T_L , the design spectral response acceleration, S_a , shall be taken as given in Eq. (11.4-6):

$$S_a = \frac{S_{D1}}{T} \quad (11.4-6)$$

- For periods greater than T_L , S_a shall be taken as given in Eq. (11.4-7):

$$S_a = \frac{S_{D1}T_L}{T^2} \quad (11.4-7)$$

where

S_{DS} = the design spectral response acceleration parameter at short periods

S_{D1} = the design spectral response acceleration parameter at a 1-s period

T = the fundamental period of the structure, s

$T_0 = 0.2(S_{D1}/S_{DS})$

$T_S = S_{D1}/S_{DS}$, and

T_L = long-period transition period(s) shown in Figs. 22-14 through 22-17.

11.4.7 Risk-Targeted Maximum Considered Earthquake (MCE_R) Response Spectrum. Where an MCE_R response spectrum is required, it shall be determined by multiplying the design response spectrum by 1.5.

11.4.8 Site-Specific Ground Motion Procedures. A site response analysis shall be performed in accordance with Section 21.1 for structures on Site Class F sites, unless exempted in accordance with Section 20.3.1. A ground motion hazard analysis shall be performed in accordance with Section 21.2 for the following:

- seismically isolated structures and structures with damping systems on sites with S_1 greater than or equal to 0.6,
- structures on Site Class E sites with S_s greater than or equal to 1.0, and,
- structures on Site Class D and E sites with S_1 greater than or equal to 0.2.

EXCEPTION: A ground motion hazard analysis is not required for structures other than seismically isolated structures and structures with damping systems where:

- Structures on Site Class E sites with S_s greater than or equal to 1.0, provided the site coefficient F_a is taken as equal to that of Site Class C.
- Structures on Site Class D sites with S_1 greater than or equal to 0.2, provided the value of the seismic response coefficient C_s is determined by Eq. (12.8-2) for values of $T \leq 1.5T_s$ and taken as equal to 1.5 times the value computed in accordance with either Eq. (12.8-3) for $T_L \geq T > 1.5T_s$ or Eq. (12.8-4) for $T > T_L$.
- Structures on Site Class E sites with S_1 greater than or equal to 0.2, provided that T is less than or equal to T_s and the equivalent static force procedure is used for design.

It shall be permitted to perform a site response analysis in accordance with Section 21.1 and/or a ground motion hazard analysis in accordance with Section 21.2 to determine ground motions for any structure.

When the procedures of either Section 21.1 or 21.2 are used, the design response spectrum shall be determined in accordance with Section 21.3, the design acceleration parameters shall be determined in accordance with Section 21.4, and, if required, the MCE_G peak ground acceleration parameter shall be determined in accordance with Section 21.5.

11.5 IMPORTANCE FACTOR AND RISK CATEGORY

11.5.1 Importance Factor. An Importance Factor, I_e , shall be assigned to each structure in accordance with Table 1.5-2.

11.5.2 Protected Access for Risk Category IV. Where operational access to a Risk Category IV structure is required through an adjacent structure, the adjacent structure shall conform to the requirements for Risk Category IV structures. Where operational access is less than 10 ft (3.048 m) from an interior lot line or another structure on the same lot, protection from potential falling debris from adjacent structures shall be provided by the owner of the Risk Category IV structure.

11.6 SEISMIC DESIGN CATEGORY

Structures shall be assigned a Seismic Design Category in accordance with this section.

Risk Category I, II, or III structures located where the mapped spectral response acceleration parameter at 1-s period, S_1 , is greater than or equal to 0.75 shall be assigned to Seismic Design Category E. Risk Category IV structures located where the mapped spectral response acceleration parameter at 1-s period, S_1 , is greater than or equal to 0.75 shall be assigned to Seismic Design Category F. All other structures shall be assigned to a Seismic Design Category based on their Risk Category and the design spectral response acceleration parameters, S_{DS} and S_{D1} , determined in accordance with Section 11.4.5. Each building and structure shall be assigned to the more severe Seismic Design Category in accordance with Table 11.6-1 or 11.6-2, irrespective

TABLE 11.6-1 Seismic Design Category Based on Short-Period Response Acceleration Parameter

Value of S_{DS}	Risk Category	
	I or II or III	IV
$S_{DS} < 0.167$	A	A
$0.167 \leq S_{DS} < 0.33$	B	C
$0.33 \leq S_{DS} < 0.50$	C	D
$0.50 \leq S_{DS}$	D	D

TABLE 11.6-2 Seismic Design Category Based on 1-s Period Response Acceleration Parameter

Value of S_{D1}	Risk Category	
	I or II or III	IV
$S_{D1} < 0.067$	A	A
$0.067 \leq S_{D1} < 0.133$	B	C
$0.133 \leq S_{D1} < 0.20$	C	D
$0.20 \leq S_{D1}$	D	D

of the fundamental period of vibration of the structure, T . The provisions in Chapter 19 shall not be used to modify the spectral response acceleration parameters for determining Seismic Design Category.

Where S_1 is less than 0.75, the Seismic Design Category is permitted to be determined from Table 11.6-1 alone where all of the following apply:

1. In each of the two orthogonal directions, the approximate fundamental period of the structure, T_a , determined in accordance with Section 12.8.2.1 is less than $0.8T_s$, where T_s is determined in accordance with Section 11.4.6.
2. In each of two orthogonal directions, the fundamental period of the structure used to calculate the story drift is less than T_s .
3. Eq. (12.8-2) is used to determine the seismic response coefficient C_s .
4. The diaphragms are rigid in accordance with Section 12.3; or, for diaphragms that are not rigid, the horizontal distance between vertical elements of the seismic force-resisting system does not exceed 40 ft (12.192 m).

Where the alternate simplified design procedure of Section 12.14 is used, the Seismic Design Category is permitted to be determined from Table 11.6-1 alone, using the value of S_{DS} determined in Section 12.14.8.1, except that where S_1 is greater than or equal to 0.75, the Seismic Design Category shall be E.

11.7 DESIGN REQUIREMENTS FOR SEISMIC DESIGN CATEGORY A

Buildings and other structures assigned to Seismic Design Category A need only comply with the requirements of Section 1.4. Nonstructural components in SDC A are exempt from seismic design requirements. In addition, tanks assigned to Risk Category IV shall satisfy the freeboard requirement in Section 15.6.5.1.

11.8 GEOLOGIC HAZARDS AND GEOTECHNICAL INVESTIGATION

11.8.1 Site Limitation for Seismic Design Categories E and F. A structure assigned to Seismic Design Category E or F shall not be located where a known potential exists for an active fault to cause rupture of the ground surface at the structure.

11.8.2 Geotechnical Investigation Report Requirements for Seismic Design Categories C through F. A geotechnical investigation report shall be provided for a structure assigned to Seismic Design Category C, D, E, or F in accordance with this section. An investigation shall be conducted, and a report shall be submitted that includes an evaluation of the following potential geologic and seismic hazards:

- a. Slope instability,
- b. Liquefaction,
- c. Total and differential settlement, and
- d. Surface displacement caused by faulting or seismically induced lateral spreading or lateral flow.

The report shall contain recommendations for foundation designs or other measures to mitigate the effects of the previously mentioned hazards.

EXCEPTION: Where approved by the authority having jurisdiction, a site-specific geotechnical report is not required

where prior evaluations of nearby sites with similar soil conditions provide direction relative to the proposed construction.

11.8.3 Additional Geotechnical Investigation Report Requirements for Seismic Design Categories D through F.

The geotechnical investigation report for a structure assigned to Seismic Design Category D, E, or F shall include all of the following, as applicable:

1. The determination of dynamic seismic lateral earth pressures on basement and retaining walls caused by design earthquake ground motions.
2. The potential for liquefaction and soil strength loss evaluated for site peak ground acceleration, earthquake magnitude, and source characteristics consistent with the MCE_G peak ground acceleration. Peak ground acceleration shall be determined based on either (1) a site-specific study taking into account soil amplification effects as specified in Section 11.4.8 or (2) the peak ground acceleration PGA_M , from Eq. (11.8-1).

$$PGA_M = F_{PGA} \cdot PGA \quad (11.8-1)$$

where

PGA_M = MCE_G peak ground acceleration adjusted for site class effects.

PGA = Mapped MCE_G peak ground acceleration shown in Figs. 22-9 through 22-13.

F_{PGA} = Site coefficient from Table 11.8-1.

where Site Class D is selected as the default site class per Section 11.4.3, the value of F_{PGA} shall not be less than 1.2.

3. Assessment of potential consequences of liquefaction and soil strength loss, including, but not limited to, estimation of total and differential settlement, lateral soil movement, lateral soil loads on foundations, reduction in foundation soil-bearing capacity and lateral soil reaction, soil down-drag and reduction in axial and lateral soil reaction for pile foundations, increases in soil lateral pressures on retaining walls, and flotation of buried structures.
4. Discussion of mitigation measures such as, but not limited to, selection of appropriate foundation type and depths, selection of appropriate structural systems to accommodate anticipated displacements and forces, ground stabilization, or any combination of these measures and how they shall be considered in the design of the structure.

TABLE 11.8-1 Site Coefficient F_{PGA}

Site Class	Mapped Maximum Considered Geometric Mean (MCE_G) Peak Ground Acceleration, PGA					
	$PGA \leq 0.1$	$PGA = 0.2$	$PGA = 0.3$	$PGA = 0.4$	$PGA = 0.5$	$PGA \geq 0.6$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.9	0.9	0.9	0.9	0.9	0.9
C	1.3	1.2	1.2	1.2	1.2	1.2
D	1.6	1.4	1.3	1.2	1.1	1.1
E	2.4	1.9	1.6	1.4	1.2	1.1
F	See Section 11.4.8					

Note: Use straight-line interpolation for intermediate values of PGA.

11.9 VERTICAL GROUND MOTIONS FOR SEISMIC DESIGN

11.9.1 General. If the option to incorporate the effects of vertical seismic ground motions is exercised in lieu of the requirements of Section 12.4.2.2, the requirements of this section are permitted to be used in the determination of the vertical design earthquake ground motions. The requirements of Section 11.9 shall only apply to structures in Seismic Design Categories C, D, E, and F.

11.9.2 MCE_R Vertical Response Spectrum. Where a vertical response spectrum is required by this standard and site-specific procedures are not used, the MCE_R vertical response spectral acceleration, S_{aMv} , shall be developed as follows:

1. For vertical periods less than or equal to 0.025 s, S_{aMv} shall be determined in accordance with Eq. (11.9-1) as follows:

$$S_{aMv} = 0.3C_v S_{MS} \quad (11.9-1)$$

2. For vertical periods greater than 0.025 s and less than or equal to 0.05 s, S_{aMv} shall be determined in accordance with Eq. (11.9-2) as follows:

$$S_{aMv} = 20C_v S_{MS} (T_v - 0.025) + 0.3C_v S_{MS} \quad (11.9-2)$$

3. For vertical periods greater than 0.05 s and less than or equal to 0.15 s, S_{aMv} shall be determined in accordance with Eq. (11.9-3) as follows:

$$S_{aMv} = 0.8C_v S_{MS} \quad (11.9-3)$$

4. For vertical periods greater than 0.15 s and less than or equal to 2.0 s, S_{aMv} shall be determined in accordance with Eq. (11.9-4) as follows:

$$S_{aMv} = 0.8C_v S_{MS} \left(\frac{0.15}{T_v} \right)^{0.75} \quad (11.9-4)$$

where

C_v = is defined in terms of S_S in Table 11.9-1,
 S_{MS} = the MCE_R spectral response acceleration parameter at short periods, and
 T_v = the vertical period of vibration.

TABLE 11.9-1 Values of Vertical Coefficient C_v

Mapped MCE_R Spectral Response Parameter at Short Periods ^a	Site Class A, B	Site Class C	Site Class D, E, F
$S_S \geq 2.0$	0.9	1.3	1.5
$S_S = 1.0$	0.9	1.1	1.3
$S_S = 0.6$	0.9	1.0	1.1
$S_S = 0.3$	0.8	0.8	0.9
$S_S \leq 0.2$	0.7	0.7	0.7

^aUse straight-line interpolation for intermediate values of S_S .

S_{aMv} shall not be less than one-half of the corresponding S_{aM} for horizontal components determined in accordance with the general or site-specific procedures of Section 11.4 or Chapter 21, respectively.

For vertical periods greater than 2.0 s, S_{aMv} shall be developed from a site-specific procedure; however, the resulting ordinate of S_{aMv} shall not be less than one-half of the corresponding S_a for horizontal components determined in accordance with the general or site-specific procedures of Section 11.4 or Chapter 21, respectively.

In lieu of using the above procedure, a site-specific study is permitted to be performed to obtain S_{aMv} at vertical periods less than or equal to 2.0 s, but the value so determined shall not be less than 80% of the S_{aMv} value determined from Eqs. (11.9-1) through (11.9-4).

11.9.3 Design Vertical Response Spectrum. The design vertical response spectral acceleration, S_{av} , shall be taken as two-thirds of the value of S_{aMv} determined in Section 11.9.2.

11.10 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

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